Label Quality

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

Teach evidence-based labeling and bounded agreement.

Use complete descriptions and checked decisions; collect actual independent annotation outcomes.

Keep one habit: Keep original labels, review decisions, rule versions and subset denominators distinct.

MAIN LESSON · SECTION 1

Prepare the connected experiment

> Keep prior exposure, shared practice and independent evidence distinct.

Goal and main teaching plan

Primary target: Write an observable **label rule** with boundaries and a review option

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Materials and prerequisite

Teach two 45-minute sessions using invented scene descriptions. The target is applying a stated **observation** rule, not certifying real robot safety. Keep original independent labels separate from reviewed decisions.

Exact **agreement** includes review in the main exercise; review is a workflow status, not a physical lane class. Avoid calling **annotator agreement** accuracy.

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Unknown means the measurement is unavailable. Zero is a recorded value permitted by that schema; replacing unknown with zero invents **evidence**.

Printed lesson and workbook

Scene cards for the four worked examples and five independent cases

Optional Python 3 editor with no extra packages

Blank decision log with item originals **evidence** rule decision reviewer

Opening retrieval

Without opening the old lesson, explain one key idea from Inspect Before Modeling. Give an example and a mistake that the idea helps you avoid.

Look for: Inspect row meaning missing values and repeated records

Reference example, not the only acceptable wording: One row is intended to represent one indoor practice trip.

The source is a teacher-authored fictional export for this lesson, duration uses minutes, and None means unknown rather than zero.

Without opening the old lesson, explain one key idea from Classification or Regression?. Give an example and a mistake that the idea helps you avoid.

Look for: Identify a task from the meaning of its target

Reference example, not the only acceptable wording: Route type is classification because 1 and 2 encode categories. Whole-minute duration is regression because numerical differences measure time.

Exceeding 10 minutes is binary classification because the target is yes or no.

End goal

By the end, I can explain label quality using a traced example and a stated limitation.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Inspect Before Modeling

Classification or Regression?

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Two labelers agree on 16 of 20 leaves. Both used a rule that mistakenly calls shadows 'spots'. Compute **agreement** and explain why it does not establish correct labels.

Concept to explain

Ambiguous definitions and inconsistent **annotation** can make the learning target unreliable; annotator agreement alone is not a universal numerical ceiling on model performance.

Term	Meaning
annotation	human assignment of a label
label rule	observable criterion for assigning a class
agreement	fraction of labels that match
adjudication	resolve disagreements using evidence

Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Readiness answer

End goal for the pictorial case

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Model rule use before showing an agreement score

> A review status belongs to the workflow, not to the lane’s physical state.

Complete corridor rule and examples

A model learns from examples with recorded targets. If two people interpret a target differently, their labels can mix different tasks.

This week you will write a **label rule**, compare independent labels and keep a record of disputed cases. We use invented corridor descriptions, not real images or a system for controlling a robot.

The classroom task is to label whether a marked practice lane is blocked in a supplied description.

Rule v1 says blocked means an object touches or crosses either lane boundary or lies inside the lane; clear means all objects are outside both boundaries and the entire lane is visible.

Use review if visibility or position is insufficient to decide. Review is a workflow status for a person to inspect, not a third physical condition of the lane.

Description	Rule v1 result	Reason
A box lies inside a fully visible lane	blocked	Inside satisfies the rule
A wheel touches the boundary line	blocked	Touching is explicitly included

Description	Rule v1 result	Reason
All objects are outside and the lane is fully visible	clear	Both clear conditions hold
A curtain hides part of the lane	review	Hidden space cannot be checked

These are exercise definitions, not real robot safety rules. Do not infer that a lane is physically safe from this label. The label describes only the supplied **evidence** under this version of the rule.

Original paired-count reasoning

Two **annotator**s used an earlier vague instruction, mark an obstructed lane, on four invented items. Their labels are shown in the code. A and D match; B and C differ.

Compare the same item IDs, not merely the first row in two differently ordered files.

The **output** is 2 4: two matching labels among four paired items, so observed exact **agreement** is 2/4 or 50 percent. Here review is included as a label value in the comparison: review versus clear is a dis**agreement**.

This simple count is not model accuracy and does not establish which labels are correct.

The code assumes one valid pair per unique item ID and allowed label strings. It does not validate IDs or spelling. Use a nonempty paired set before calculating a fraction; zero paired items gives no **agreement** estimate.

Term	Meaning
Label rule	A written definition for assigning an outcome to an example
Annotator	A person who assigns labels using a stated rule
Agreement	How often annotators assign the same label to the same items
Adjudication	Reviewing evidence and rules to resolve a disputed label or retain uncertainty

Review B and C without overwriting labels

For B, the full lane is visible and a wheel touches its boundary. One **annotator** treated touching as clear; the other treated it as blocked. Rule v1 explicitly includes touching, so the reviewed label is blocked.

Preserve both original labels and record the item, rule version, **evidence**, decision and reviewer. Do not silently overwrite the originals.

For C, a curtain hides part of the lane. One **annotator** guessed clear; the other requested review. Rule v1 requires full visibility for clear. The justified status remains review until sufficient **evidence** is available.

More votes cannot reveal the hidden area.

> **Using rule v1, explain the decisions for B and C. If you exclude C from the original comparison, what is agreement among the remaining pairs? Why must you still report the exclusion?**

A decision log could say: B, original clear/blocked, wheel touches line, v1, blocked after review by the teacher. For C it could say: original review/clear, lane partly hidden, v1, unresolved review.

Add the labeling method and rule version to the dataset card.

Read separate leaf facts and limitations

Write a **label rule** that resolves the visible boundary case.

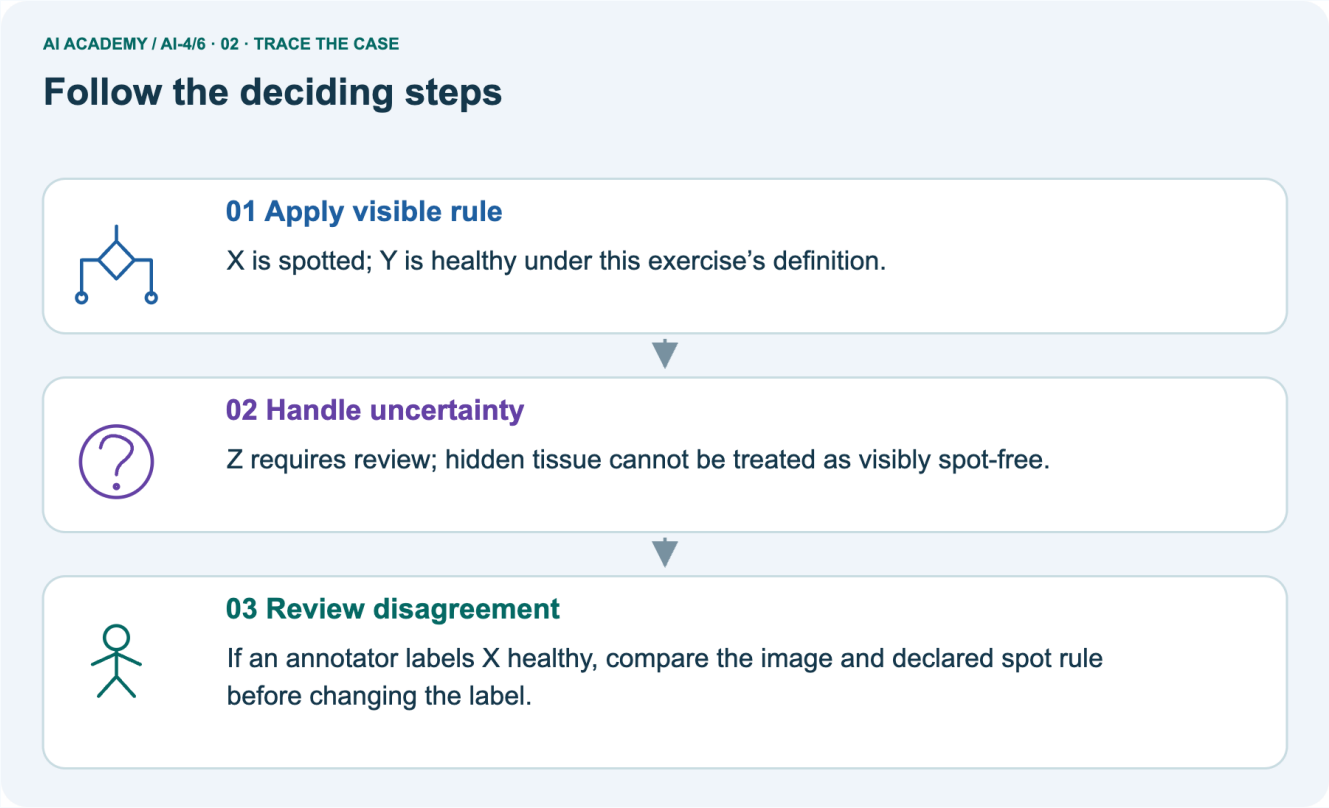
Toy spotted-leaf rule: at least one fully visible brown spot means spotted; occluded leaves require review. X has one visible spot; Y has none; Z is half hidden.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The leaf picture uses a separate toy rule. Healthy here means no visible brown spots on a fully visible leaf under this exercise, not actual plant health. A new fully visible image supplies new evidence; it does not make the older hidden image informative. For corridor pairs A–D,

excluding C leaves A,B,D with two original matches out of three; reviewed labels do not replace the original comparison.

Trace rule, uncertainty and new observation



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Apply visible rule	X is spotted; Y is healthy under this exercise’s definition.
2. Handle uncertainty	Z requires review; hidden tissue cannot be treated as visibly spot-free.
3. Review disagreement	If an annotator labels X healthy, compare the image and declared spot rule before changing the label.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Read the picture in order. If an **annotator** labels X healthy, compare the image and declared spot rule before changing the label. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Visible brown spots are a simplified classroom target, not a diagnosis of plant disease.

Change: Z receives a new fully visible image with no brown spots.

Prediction: Label the new **observation** healthy under the toy rule.

Reason: The missing visual **evidence** is now supplied; the earlier uncertain image was not proof of health.

▼ **Earlier wording, alternatives and picture transcripts**

Explain the mechanism

Trace the information, fitted behavior, and evaluation proof separately.

QUESTION Measure **annotator agreement** and revise the labeling guide.

MODEL IDEA Ambiguous definitions and inconsistent **annotation** can make the learning target unreliable; annotator agreement alone is not a universal numerical ceiling on model performance.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Two **annotator**s label 20 leaves and agree on 16. Observed **agreement** = 16 ÷ 20 = **80%**.

Draw a fresh data → model → **prediction** → evaluation trace:

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

STEP 1 Two **annotator**s label 20 leaves and agree on 16.

STEP 2 Observed **agreement** = 16 ÷ 20 = **80%**.

STEP 3 Most dis**agreement**s concern tiny spots near the boundary.

STEP 4 Define a minimum spot rule, relabel a fixed subset, and report the new **agreement**.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

Concrete teaching case

Two **annotator**s disagree on three of ten leaves, especially mild spotting.

Improve label quality.

Exact explanation to model

Define spotting operationally, inspect dis**agreement**, use appropriate **adjudication** and document ambiguity. **Agreement** is 7/10, but **agreement** is not proof of truth.

Explain the added worked example

Key terms

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

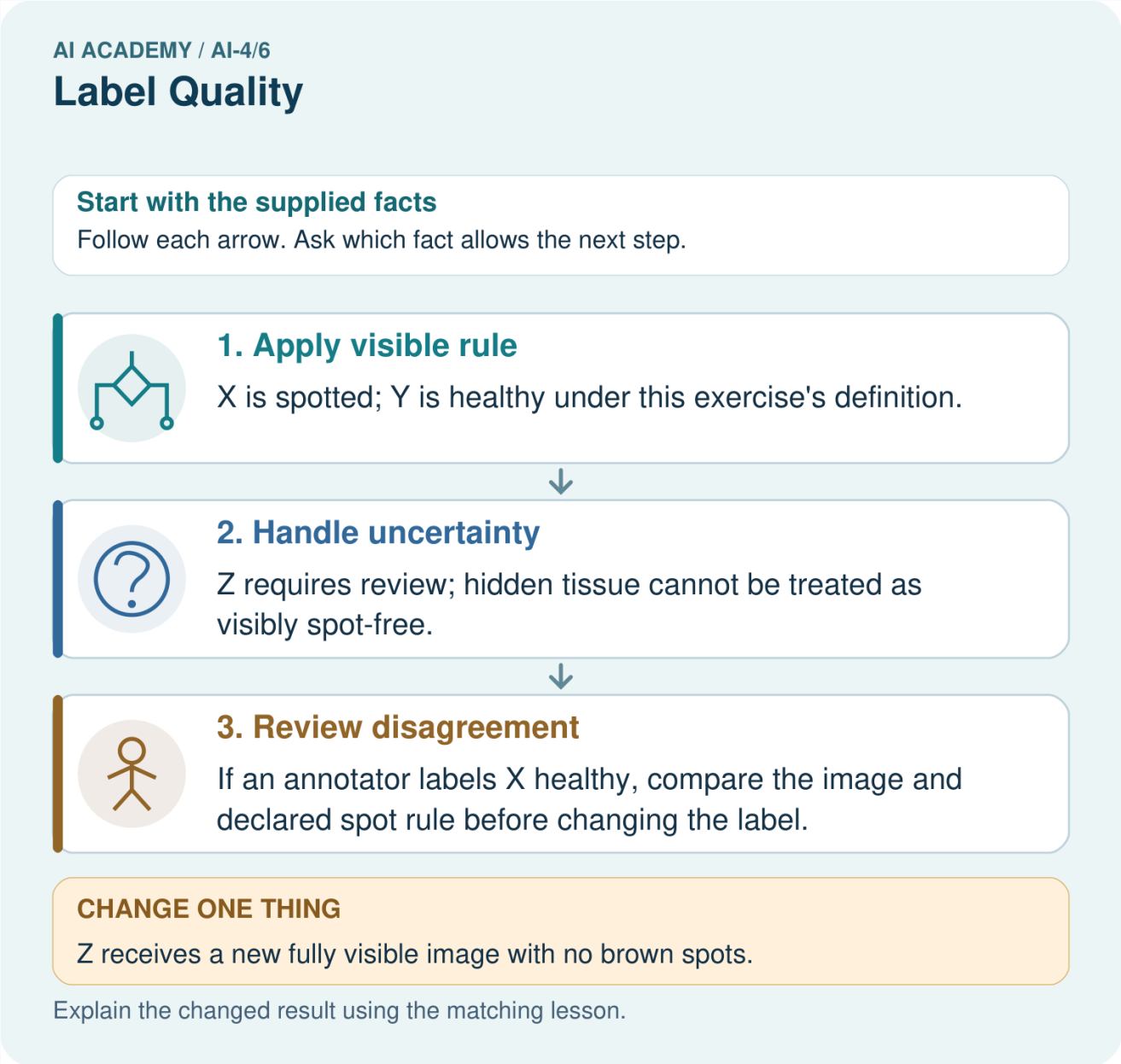
These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Label Quality

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Guide the calculation, then remove help

> Collect each learner's prediction before showing its key.

Current completion task and checked key

Completion task: Explain the **output** 2 4 for A through D. State the fraction and percent, whether review is included, and why this is not model accuracy.

Teacher check: Two of four matched items have identical **annotator** labels: 2/4 or 50 percent. Review is included in this exact comparison.

Neither **annotator** is established as a correct reference, and no model **prediction**s are being scored.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for Label definitions and dis**agreement**. Proposed explanation: "**Agreement** is accuracy". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: **Agreement** compares **annotator**s. Accuracy would require **prediction**s and an appropriate reference; shared mistakes can still match.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial question and changed-case reasoning

Why should two **annotator**s not vote away Z's missing **evidence**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Agreement or a majority cannot reveal hidden tissue. Seek an adequate **observation** and preserve the review status until then. Changed-case check: Label the new **observation** healthy under the toy rule.

The missing visual **evidence** is now supplied; the earlier uncertain image was not proof of health.

Changed-case reasoning: Label the new **observation** healthy under the toy rule. The missing visual **evidence** is now supplied; the earlier uncertain image was not proof of health.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Keep each learner explaining

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

▼ Earlier wording, alternatives and picture transcripts

Compact alternative: this two-session sequence covers selected core tasks. Use the three-session weekly plan for the full lesson, independent transfer, feedback and project work; allow extra time for extensions.

Demonstrate and guide practice

Board sequence

Write the given facts: Two **annotator**s disagree on three of ten leaves, especially mild spotting.

Write the question to explain: Improve label quality.

Reveal after **prediction**s: Define spotting operationally, inspect dis**agreement**, use appropriate **adjudication** and document ambiguity. **Agreement** is 7/10, but **agreement** is not proof of truth.

Use the earlier transfer question as additional practice. Reserve the five new P–T annotation pairs for the first individual attempt and record support used.

Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

Model these guided questions

Guided: B is blocked because touching counts; C stays review because the lane is partly hidden. Excluding C leaves the original pairs A, B and D: 2 matches out of 3, about 66.7 percent.

Report the exclusion because this is a different subset from the four-item comparison, not newly correct labels. Exit: perfect **agreement** establishes matching labels on these items only. Both people can share a rule error.

Check their labels against the boundary **evidence** and v1 before making a correctness claim.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve unknown versus zero. Ask why a label needs an explicit meaning before it becomes training data.
7 to 18 minutes	Read v1. Apply inside, boundary, outside and hidden examples. Establish that these are fictional descriptions, not robot safety advice.
18 to 32 minutes	Have pairs label fresh verbal examples independently before talking. Trace the supplied four-item code and complete W1 and W2.
32 to 45 minutes	Compare B and C using evidence. Complete the guided denominator question and preserve the original pair list.
Session 2 start to 8 minutes	Retrieve agreement versus correctness. Show a shared mistake on a touching-boundary example.
8 to 20 minutes	Model a decision-log entry without erasing original labels. Complete W3 and discuss a genuine rule revision.
20 to 35 minutes	Collect W4 independently before support. Require both denominators, rule-based decisions and the unresolved review case.
35 to 45 minutes	Use W5 and the exit response. After feedback, offer the fresh retry and record any support used.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Keep predictions, observed or supplied outcomes and unmeasured cases separate.

Supplied inputs and checked guidance

Label Lab v1 expected card statuses: L01 blocked,L02 blocked,L03 clear,L04 review,L05 blocked,L06 clear,L07 review,L08 blocked,L09 blocked,L10 clear,L11 review,L12 blocked,L13 clear,L14 review,L15 blocked,L16 blocked,L17 clear,L18 review,L19 blocked,L20 clear. Actual annotator agreement must be collected, not invented. Changed L03 with hidden contents requires review. Shared mistakes can match perfectly.

Selected P–T original agreement3/5=60%; excluding R gives2/4=50%; reviewed decisions blocked,blocked,review,clear,blocked. Clarify that original pairs refers to P–T here. Main A–D exclusion instead gives2/3.

Predict, investigate and record

1. 1. Write the hypothesis and expected **evidence** before running.
2. 2. Label Lab: independently label twenty cards, compute **agreement**, cluster dis**agreement**s, revise one rule, and repeat.
3. 3. Run one normal, one boundary, one missing, and one shifted case.
4. 4. Calculate the metric from raw counts or row-level errors.
5. 5. Interpret what changed, what did not, and what remains unsafe or unknown.

Keep the experiment ledger

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose the current misconceptions

Agreement is accuracy: **Agreement** compares **annotator**s. Accuracy would require **prediction**s and an appropriate reference; shared mistakes can still match.

Review means clear: Review preserves insufficient **evidence**. It is neither a negative label nor proof of a physical condition.

Voting removes uncertainty: Votes may select a decision policy but cannot create missing **evidence**. Preserve unresolved cases and document the policy.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Keep the test conditions fixed, record every result, and change only one planned factor.

Make the lab runnable

Use the supplied complete investigation for shared practice. Reserve the five new P–T annotation pairs for the individual first attempt before hints or feedback; earlier transfer questions remain additional practice.

Diagnose and reteach

MISCONCEPTION TO REPAIR The model can learn the correct concept even when people label the same example differently.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the selected experiment transfer

> Keep this key until after a first attempt; record any help and prior exposure.

Selected independent task and checked key

Selected case: New pairs are P blocked/blocked, Q clear/blocked, R review/review, S clear/clear, T blocked/clear.

P has an object inside; Q touches the line; R is partly hidden; S has all objects outside a fully visible lane; T has an object inside. Count **agreement** including review. Apply v1 to all five.

Then report **agreement** on the original pairs excluding any review status, with its denominator and limitation.

Three of five pairs match, or 60 percent. Reviewed results are P blocked, Q blocked, R review, S clear, T blocked. Excluding R leaves two matches among four pairs, or 50 percent.

The subset omits the uncertain case and does not establish correctness or general reliability. Preserve original labels separately from reviewed decisions.

Judge trace, scope and support

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Use the boundary and review rules

Under rule v1, label a wheel touching the boundary, an object entirely outside a fully visible lane, and a partly hidden lane. Explain the **evidence** for each.

Touching is blocked. An object entirely outside is clear only when all objects are outside and the entire lane is visible.

A partly hidden lane requires review because clear cannot be established from the available description.

W2 Interpret agreement

Explain the **output** 2 4 for A through D. State the fraction and percent, whether review is included, and why this is not model accuracy.

Two of four matched items have identical **annotator** labels: 2/4 or 50 percent. Review is included in this exact comparison. Neither **annotator** is established as a correct reference, and no model **prediction**s are being scored.

W3 Write a decision log

For B and C, record original labels, visible **evidence**, rule version and reviewed decision. Explain why the original labels should remain available.

B: clear/blocked, wheel touches boundary, v1, blocked. C: review/clear, partly hidden lane, v1, unresolved review. Record the reviewer as well.

Originals preserve the independent dis**agreement** and let another person audit how a decision was made.

W5 Draft and test a label rule

Write a short v1 labeling instruction covering blocked, clear, the touching boundary and review. Add one boundary example. Explain why three people voting clear cannot settle a hidden lane.

A suitable instruction specifies the marked lane: inside or touching either boundary means blocked; clear requires all objects outside and full visibility; insufficient position or visibility means review.

A wheel touching the line is blocked. Votes supply no missing visual **evidence**, so a hidden lane remains review under v1.

▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

Worked case answer

Define spotting operationally, inspect dis**agreement**, use appropriate **adjudication** and document ambiguity. **Agreement** is 7/10, but **agreement** is not proof of truth.

Additional transfer practice question

Two labelers assign different species to the same image. Should you silently pick the label that helps the model score? Describe a better **process**.

Reasoned practice answer

Review the labeling rule and **evidence**, resolve or mark uncertainty, and document the decision independently of model performance.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “The most useful label is whichever makes evaluation accuracy highest.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Agreement=80%. Shared misunderstanding can produce **agreement** on wrong labels. Clarify spotting criteria, inspect disputed and agreed examples, use appropriate **adjudication** and record ambiguity.

Do not treat majority **agreement** alone as ground truth.

Explanations for the additional workbook examples

W1 Use the boundary and review rules

W2 Interpret agreement

W3 Write a decision log

W4 Apply the process independently

New pairs are P blocked/blocked, Q clear/blocked, R review/review, S clear/clear, T blocked/clear.

P has an object inside; Q touches the line; R is partly hidden; S has all objects outside a fully visible lane; T has an object inside. Count **agreement** including review. Apply v1 to all five.

Then report **agreement** on the original pairs excluding any review status, with its denominator and limitation.

Three of five pairs match, or 60 percent. Reviewed results are P blocked, Q blocked, R review, S clear, T blocked. Excluding R leaves two matches among four pairs, or 50 percent.

The subset omits the uncertain case and does not establish correctness or general reliability. Preserve original labels separately from reviewed decisions.

W5 Draft and test a label rule

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, support and revisit the experiment

> Retain failures and assess a fresh parallel case after help.

Project evidence

Write the Plant Classifier label guide with positive, negative, borderline, and not-labelable examples.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Support and selected reassessment

Use a tiny labelled dataset, explicit units and a named positive class. Stretch: require a baseline, held-out evaluation and subgroup or shifted-condition analysis with denominators.

Use lane drawings and movable object cards for touching versus outside; cover part of the drawing for review. Provide blank log headings without decisions.

Extension: construct two **annotator**s who agree perfectly while both break v1 and explain why the **agreement** fraction does not reveal the error.

Require W4 counts 3/5 including review and 2/4 excluding review, all five v1 decisions, and an explicit subset limitation.

Fresh retry: U clear/clear with all objects outside a fully visible lane; V blocked/clear with an object touching the boundary; W blocked/blocked with an object inside; X review/blocked with part of the lane hidden.

Original **agreement** is 2/4 or 50 percent; excluding X gives 2/3 or about 66.7 percent. Reviewed v1 decisions are clear, blocked, blocked, review. Require **evidence** for V and X before accepting the retry.

Delayed return

Suggested return points in this level: ai-4/7, ai-4/9, ai-4/12. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge and reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 7: Sampling and Intended Use. Retrieve today's **evidence** and identify what carries forward and what is new.

Next is Week 7: Sampling and Intended Use. Carry forward item IDs, rule versions, original labels and review status so the source and covered conditions remain traceable; related-example split design is covered later.

▼ **Earlier wording, alternatives and picture transcripts**

Connect and extend learning

Responsible Plant Classifier

Support and stretch

Exit reflection

Bridge to the next week

Support and fresh reassessment

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route without repeating the whole cycle.

Planning and assessment reference

Teaching target: Write an observable **label rule** with boundaries and a review option

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

Before class, solve the supported practice: Explain the **output** 2 4 for A through D. State the fraction and percent, whether review is included, and why this is not model accuracy.

Reasoned practice answer: Two of four matched items have identical **annotator** labels: 2/4 or 50 percent. Review is included in this exact comparison.

Neither **annotator** is established as a correct reference, and no model **prediction**s are being scored.

Check every learner before discussion: Use the worked case for Label definitions and dis**agreement**. Proposed explanation: “**Agreement** is accuracy”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: **Agreement** compares **annotator**s. Accuracy would require **prediction**s and an appropriate reference; shared mistakes can still match.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Z receives a new fully visible image with no brown spots. Require a **prediction** and a reason before comparing. Expected result: Label the new **observation** healthy under the toy rule.

Reason: The missing visual **evidence** is now supplied; the earlier uncertain image was not proof of health.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback and evidence records**

REFERENCE · SECTION 8

Reference · Feedback and evidence records

> Keep supported rehearsal separate from independent performance.

Record and repair

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

▼ **Earlier wording, alternatives and picture transcripts**

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

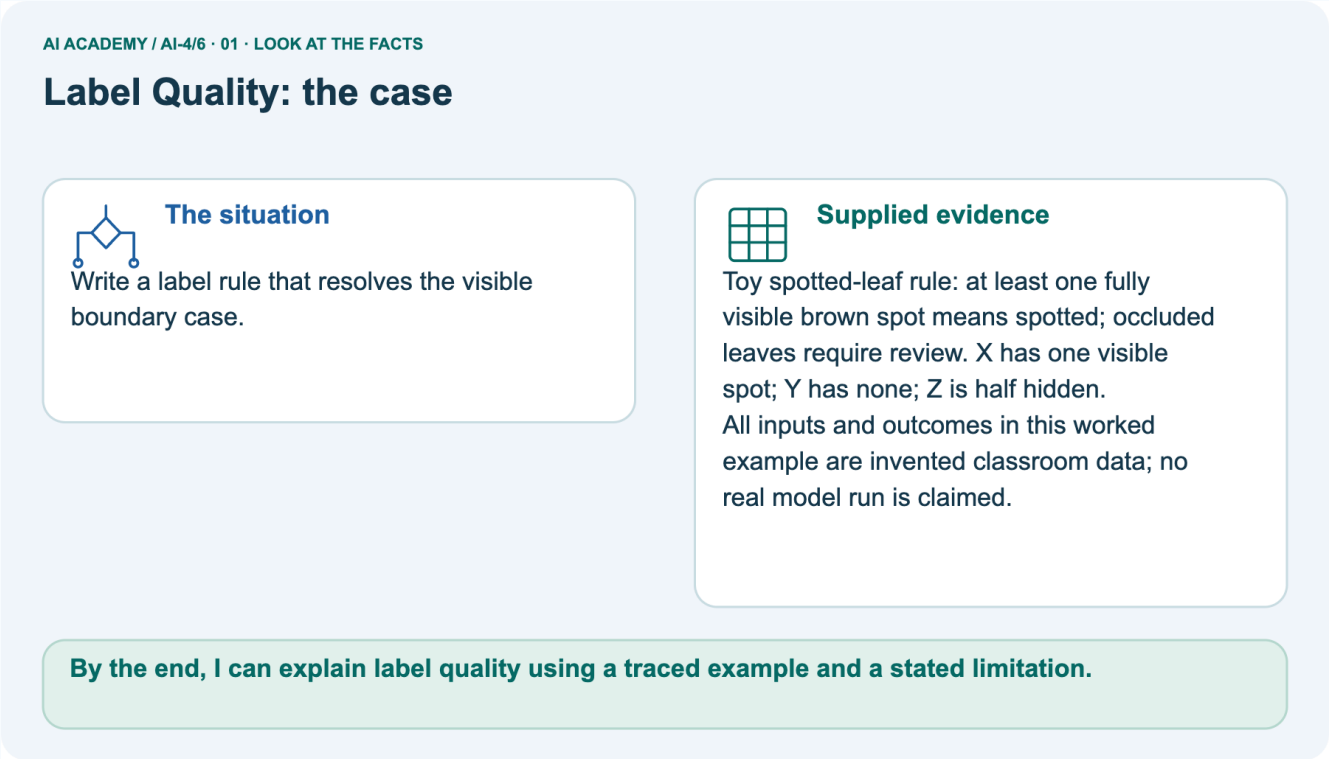
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a specific reasoning move.

01. Label Quality: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

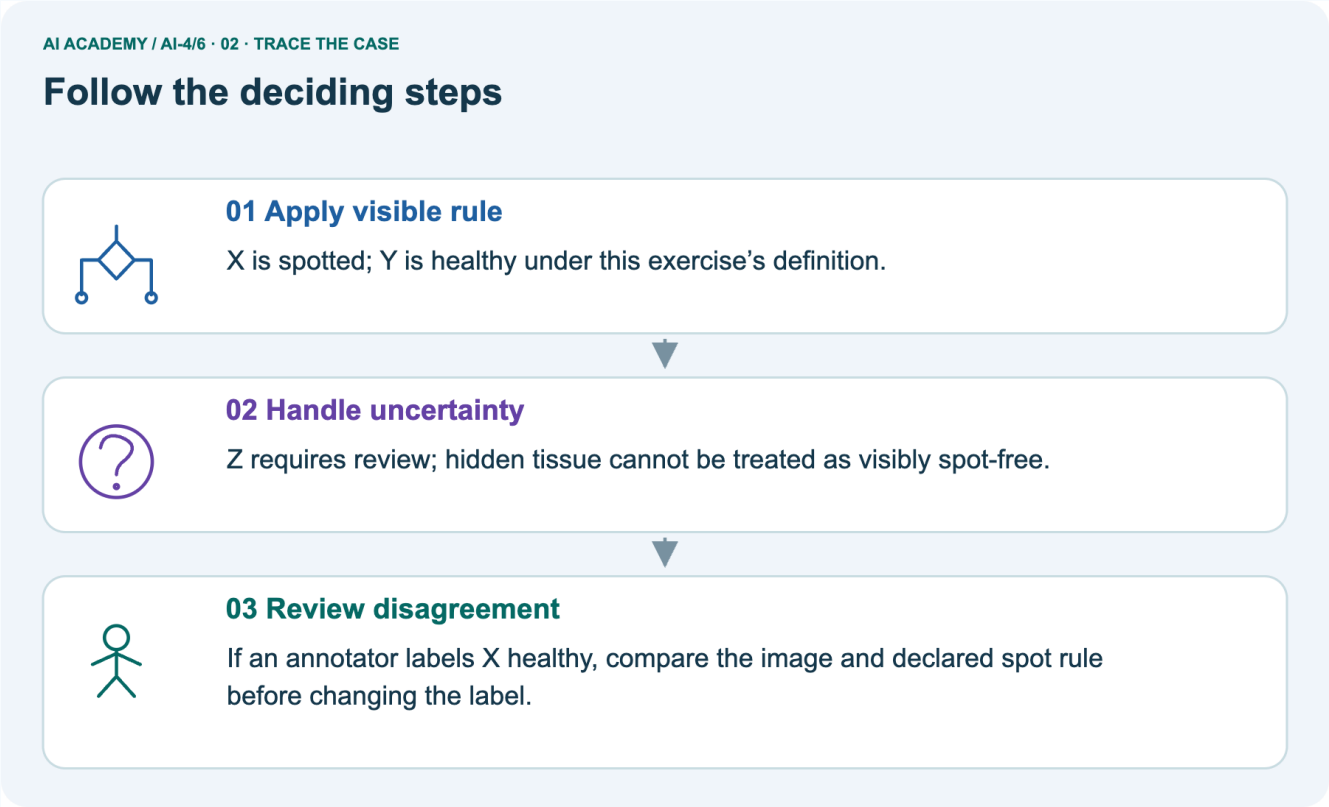
Write a **label rule** that resolves the visible boundary case. Supplied facts: Toy spotted-leaf rule: at least one fully visible brown spot means spotted; occluded leaves require review.

X has one visible spot; Y has none; Z is half hidden. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Apply visible rule: X is spotted; Y is healthy under this exercise's definition. → Handle uncertainty: Z requires review; hidden tissue cannot be treated as visibly spot-free. → Review dis**agreement**: If an **annotator** labels X healthy, compare the image and declared spot rule before changing the label.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Follow a disagreement back to evidence



Follow the labelled picture with the matching explanation. Open the vector diagram

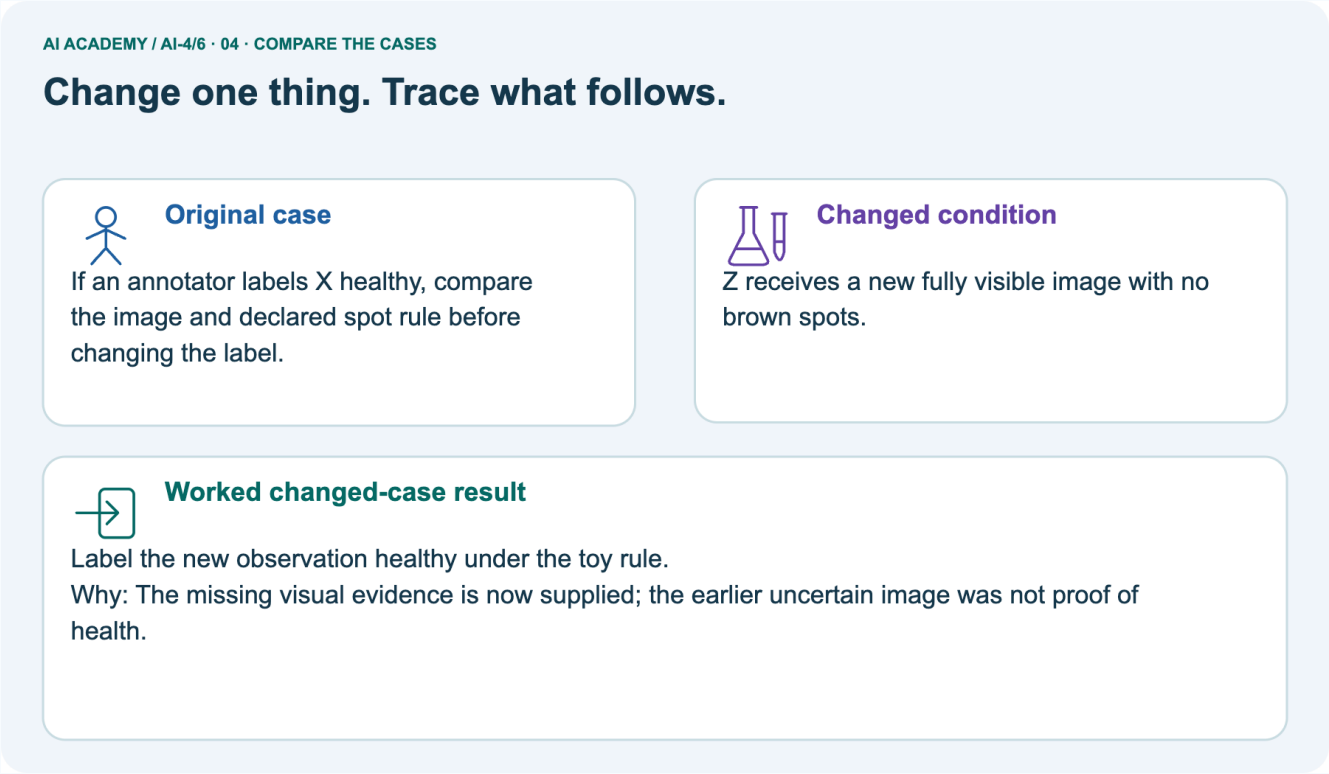
Follow a dis**agreement** back to **evidence**. B has two labels | Read boundary rule | Keep labels and reason. C includes review | Check visible **evidence** | Retain uncertainty if needed.

A and D match | Check rule as well | **Agreement** is not proof

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: If an **annotator** labels X healthy, compare the image and declared spot rule before changing the label. Changed condition: Z receives a new fully visible image with no brown spots.

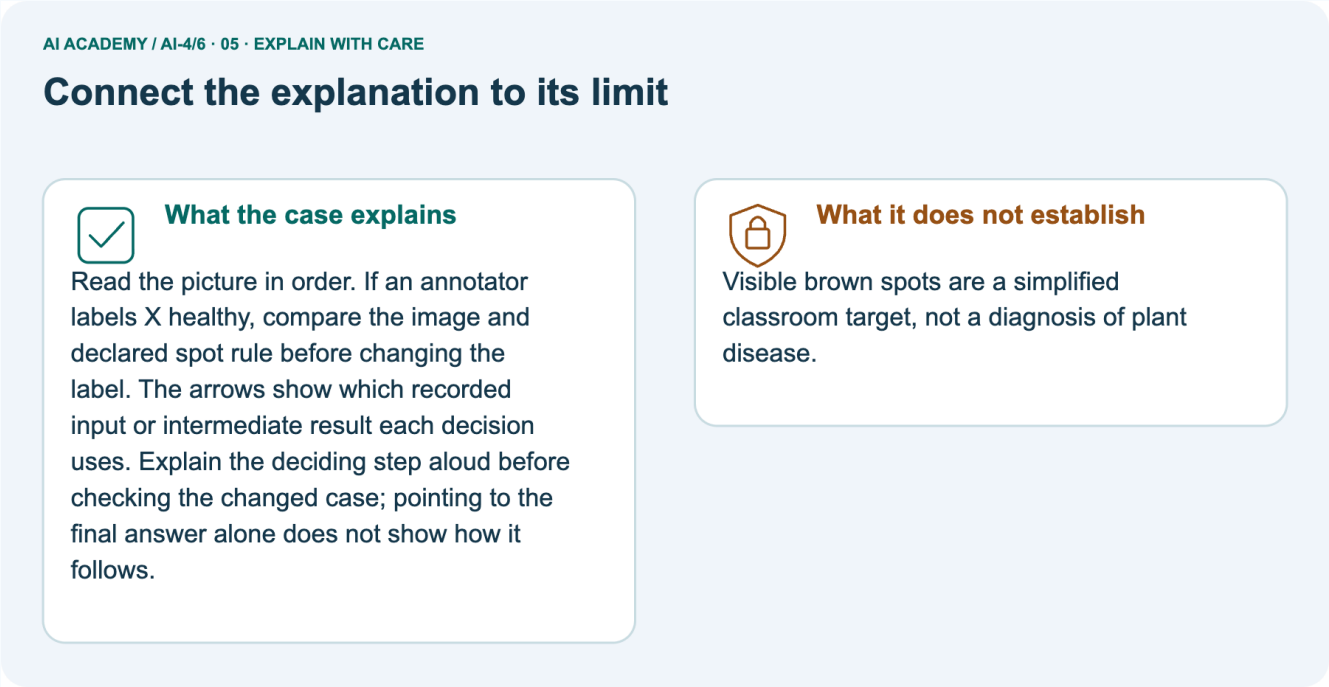
Expected result: Label the new **observation** healthy under the toy rule. Reason: The missing visual **evidence** is now supplied; the earlier uncertain image was not proof of health.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The missing visual **evidence** is now supplied; the earlier uncertain image was not proof of health.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. If an **annotator** labels X healthy, compare the image and declared spot rule before changing the label. The arrows show which recorded **input** or intermediate result each decision uses.

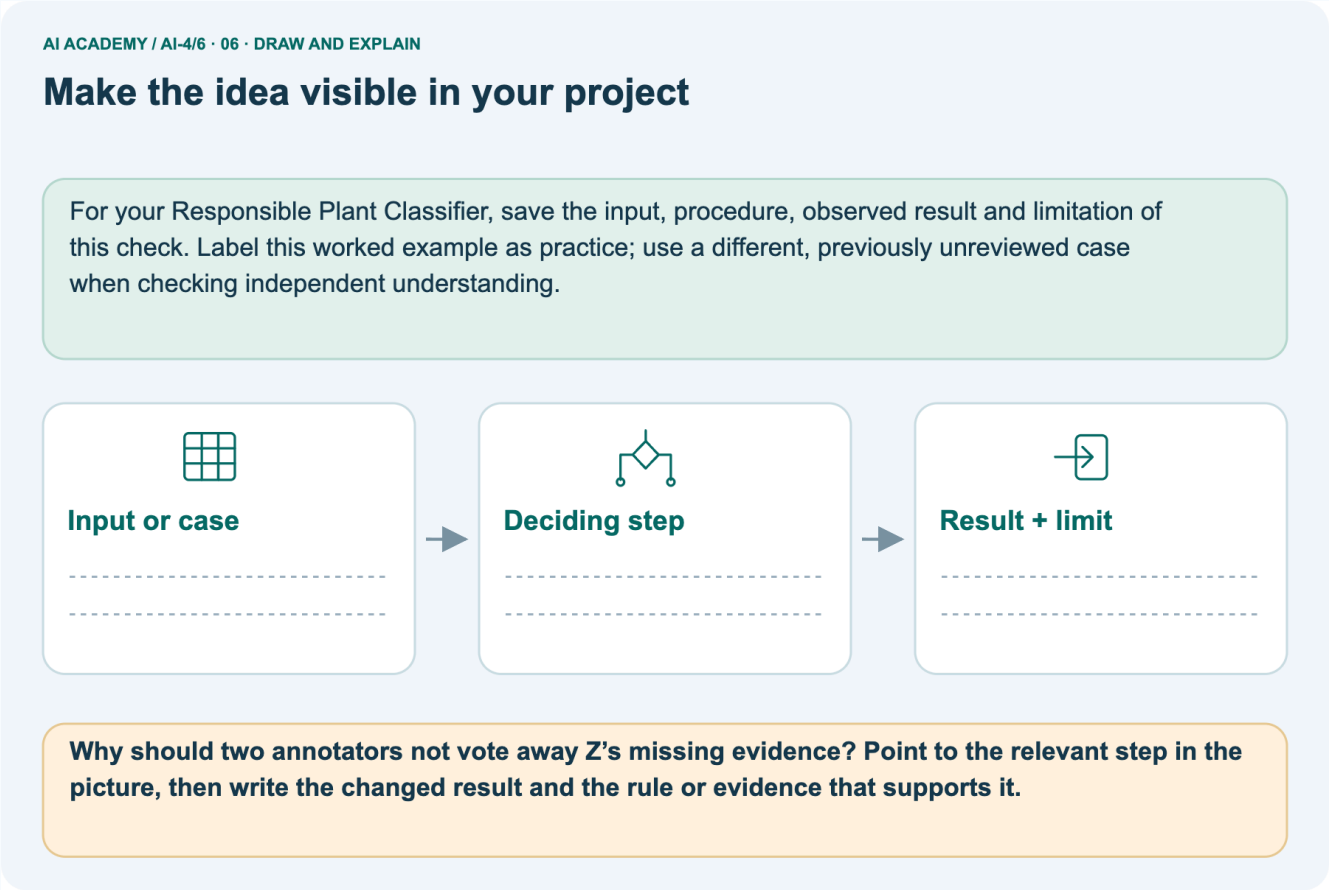
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows. Limit: Visible brown spots are a simplified classroom target, not a diagnosis of plant disease.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Visible brown spots are a simplified classroom target, not a diagnosis of plant disease.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why should two **annotator**s not vote away Z's missing **evidence**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: **Agreement** or a majority cannot reveal hidden tissue. Seek an adequate **observation** and preserve the review status until then. Changed-case check: Label the new **observation** healthy under the toy rule.

The missing visual **evidence** is now supplied; the earlier uncertain image was not proof of health.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Label Quality: the case

02. Follow the deciding steps

03. Follow a disagreement back to evidence

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.